Metaphony in Substance-Free Logical Phonology

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Abstract

We provide an overview of Substance-Free Logical Phonology (LP) and apply this model to a few case studies of synchronic metaphony and similar processes. We adopt Inkelas and colleagues' notion of *Inalterability as Prespecification* (IAP), implementing feature-filling rules with a unification operator. We show that the assumptions of LP allow us to reanalyze some cases of metaphony previously treated as morphophonological as occurring within the narrow phonology. We also generalise the IAP notion from rule targets to rule triggers.

1 Introduction

We provide an overview of Substance-Free Logical Phonology (henceforth LP) and apply this model to a few case studies of synchronic metaphony and similar processes. In LP, the purview of phonology is quite narrow, and all intrasegmental processes (i.e., featural changes affecting a single segment), are derived though two kind of rules: features can be added to a segment by rules via *unification*, and features can be removed from a segment by rules via *subtraction.* While these are the only two rule types considered in this study, insertion and deletion of full *segments*, for example, require additional machinery, as do syllabification and stress assignment. We also do not present here the LP approach to the specification of rule environments and the nature of locality, a topic of great importance in the study of metaphony and similar phenomena, though Dabbous et al. (2025) give some preliminary thoughts on this matter.

Given the fact that we reduce all intrasegmental rules to these two operations—unification rules and subtraction rules—our goal is to show that various phonological processes that go by names like assimilation, palatalization, vowel harmony, and also metaphony can be reduced to computations with the same operations and abstract representations. On one hand, this leads us to conclude that labels such as metaphony—by which we refer to root vowel alternations triggered by vowels in the following syllable—are what Chomsky (2000:8) calls "taxonomic artifacts, useful for informal description perhaps but with no theoretical standing." On the other hand, it is clear that a volume like the present one is methodologically extremely useful: the focus on metaphony serves a similar purpose to micro-variation studies in syntax that help us "to abstract from the welter of descriptive complexity certain general principles governing computation that would allow the rules of a particular language to be given in very simple forms" (op. cit.:122). We use the name *Logical Phonology* in order to reiterate the formal, substance-free nature of phonology as we conceive it, and also because we think that much of phonology can be expressed using sets and simple operations over sets. The toolkit we use here is presented in more detail by Bale and Reiss (2018), Bale et al. (2020) and Reiss (2021), and it provides us with a means of developing a constrained theory of possible phonological rules.

In section 2, we outline theoretical preliminaries, including the notion *substance-freeness*, the representation of segments as sets of valued features, the definition of the unification and subtraction operators, our assumptions about phonological acquisition, and the role of underspecification in LP. Then, we provide three case studies using the tools of LP to analyze metaphony in the Baztan dialect of Basque (section 3), Spanish (section 4), and Cervara (section 5). For LP, the fundamental problem posed by synchronic metaphony is that many of the alternations appear to show morphemic or morphosyntactic conditioning. However, we find that the assumptions of LP lead us to reanalyze certain cases of metaphony previously understood as morphophonological—in the sense that their analysis requires reference to morphemic identity and/or morphosyntactic context—as occurring within the narrow phonology.

2 Theoretical preliminaries

2.1 Substance-freeness

We assume an innate set of binary phonological features transducing from the output of the grammar to the performance systems involved in speech production and perception (Chomsky and Halle 1965, Volenec and Reiss 2018, 2019, 2025, Reiss and Volenec 2022). The model is *substance-free* in the sense that the motor and perceptual correlates of a feature are not accessible to the phonological computational system. In LP, a process changing a −LATERAL segment to +LATERAL before a +ROUND vowel has the same status as one turning a +HIGH vowel to −HIGH before a +NASAL segment, regardless of the relative phonetic plausibility of the two processes, their relative typological frequency, their putative presence or absence in children's speech, or any other factors discussed in the literature under the rubric of *markedness*.[1] Thus *Substance-free* can in some sense be understood as *markedness-free*. The two hypothetical processes above are represented differently because they are formulated with different features and coefficients, but for grammatical computations, features are arbitrary symbols as discussed by Hale and Reiss (2008).

As a point of comparison, consider assimilation, which Hyman (1975) defines as the insertion of a feature specification αF on a segment immediately adjacent to another segment specified αF.

(1) Assimilation schemata (after Hyman 1975:159):

$$X \rightarrow \{\alpha F\} \quad / \quad \_\_ \, [\alpha F]$$

$$X \rightarrow \{\alpha F\} \quad / \quad [\alpha F] \, \_\_$$

[1] We take all these facts to be external to the synchronic grammar and to reflect the *diachronic filter*—à la Ohala (2003), Hale (2003), and Blevins (2004)—or specific properties of the language acquisition device.

In autosegmental phonology, assimilation is instead conceptualised as the sharing of feature specifications via the insertion of association lines, and phonological tiers provide a more general notion of adjacency, but what these approaches all share is that they essentially reify the notion of assimilation. What LP denies is that there is any need to recognise notions like assimilation (or dissimilation) as less marked than other processes. Yes, there are rules that encode agreement between segments, for example, rules that make an obstruent agree in voicing with an obstruent to its immediate right, or that delete a glide between identical vowels, or that raise mid vowels before high vowels, and LP can easily express such rules. However, there are also rules which raise mid vowels before low vowels, before nasals, or before a word boundary. The following principle expresses the LP position in general terms.

(2) Substance-freeness of structural change (Dabbous et al. 2025): Features changed by rule application need not be present in the rule's structural environment.

This principle is a claim that proposed phonological rules need not "make sense" in featural terms. Informally, it holds that the *whatness* of a rule, which feature is being added to a segment, is logically independent of the *whereness*, the characterization of the triggering environment. This in turn echoes Chomsky and Halle's claim (1968:428) that "the phonological component requires wide latitude in the freedom to change features". Note that this principle is not an axiom of LP; it just draws attention to the fact that this stipulation is *not* part of the theory.

2.2 Segments as sets of features

For the kinds of phenomena we examine in this paper, it is sufficient to treat segments as sets of valued features. It is not necessary to concern ourselves with the association of these sets with timing slots, as one must when dealing with *segment* insertion and deletion rules, for example. Thus individual segments, just like structural changes, are denoted in the curly brackets of set theory, as in (3).[2]

(3) Segments as sets of features:

$$/\text{i}/ = \left\{\begin{array}{c} +\textsc{Syllabic} \\ -\textsc{Back} \\ -\textsc{Round} \\ +\textsc{High} \\ -\textsc{Low} \\ +\textsc{Atr} \end{array}\right\} \qquad /\text{e}/ = \left\{\begin{array}{c} +\textsc{Syllabic} \\ -\textsc{Back} \\ -\textsc{Round} \\ -\textsc{High} \\ -\textsc{Low} \\ +\textsc{Atr} \end{array}\right\}$$

LP permits underspecified segments, so a set like (4) is also a legitimate representation for a segment; it denotes a vowel that lacks a specification for the feature HIGH but which is otherwise technically identical to /i, e/. Following traditional practice, we use capital letters, like /I/, to abbreviate underspecified segments.

(4) $$/\text{I}/ = \left\{\begin{array}{c} +\textsc{Syllabic} \\ -\textsc{Back} \\ -\textsc{Round} \\ -\textsc{Low} \\ +\textsc{Atr} \end{array}\right\}$$

In set-theoretic terms, this vowel /I/ is the intersection of the vowels /i/ and /e/, each of which is a set of features. The fact that LP permits underspecification is again not an axiom; it just happens that the there is no axiom in LP requiring segments to be *complete*, i.e., fully specified for all features.

As a notational convention, LP distinguishes representation (4) from representation (5), which is a *partial description*. We use square brackets to denote natural classes; (5) is the intensional characterization of a set of segments where each segment is a set of feature specifications.

[2]We assume these features for concreteness, but the exact featural specification of these segments is not at issue.

(5) The natural class containing /i, e, I/:

$$\begin{bmatrix} +\textsc{Syllabic} \\ -\textsc{Back} \\ -\textsc{Round} \\ -\textsc{Low} \\ +\textsc{Atr} \end{bmatrix}$$

This representation is an abbreviation for unwieldy set expression (6) below.

(6) $X = \{x \mid x \supseteq \{+\textsc{Syllabic}, -\textsc{Back}, -\textsc{Round}, -\textsc{Low}, +\textsc{Atr}\}\}$

In prose, X is the set of segments whose members are all and only the segments whose featural specifications are supersets of the feature specification set {+Syllabic, −Back, −Round, −Low, +Atr}. We assume that rule targets are natural classes (e.g., "apply the rule to any labial") as are the rule environments (e.g., "apply the rule before front vowels") and we use square brackets to denote these two components. In contrast, the change specified by the rule does not refer to a set of segments, but to a set of featural changes, and therefore regular set-theoretic curly brackets are used to specify the structural change. We will encounter many examples of rules stated using these conventions below.

Given the representations for /i, e, I/ proposed above, it is possible to specify singleton natural classes of fully specified segments using the square bracket notation, attending to the distinction between natural classes and segments.

(7) Natural class versus segment:

$$\text{singleton class \{i\}:} \quad \begin{bmatrix} +\textsc{Syllabic} \\ -\textsc{Back} \\ -\textsc{Round} \\ +\textsc{High} \\ -\textsc{Low} \\ +\textsc{Atr} \end{bmatrix} \qquad \text{segment /i/:} \quad \begin{Bmatrix} +\textsc{Syllabic} \\ -\textsc{Back} \\ -\textsc{Round} \\ +\textsc{High} \\ -\textsc{Low} \\ +\textsc{Atr} \end{Bmatrix}$$

Similarly, one can denote the singleton natural class {e} by changing +High to −High on the left side of (7). However, it is *not* possible in this system to construct a natural class containing only the segment /I/. Informally, this is because the relevant set is a proper superset of the sets corresponding to /i/ and to /e/. In general, given segments p and q where p is underspecified relative to q (i.e., $p \subset q$), there is no way to construct a natural class containing p but not not also q. In the case of (5), /I/ contains the features of the natural class representation in (5), but so do /i, e/, because they are supersets of /I/.

2.3 Unification and subtraction

The focus of much work in LP is the development and application of a mechanism to overcome the apparent representational limitation pointed out above. How can feature-filling processes be formulated if it is not possible to refer exclusively to underspecified segments? LP is not the first theory to consider this issue. For example, Lees (1961:12–14) discusses it in the context of Turkish vowel harmony and proposes a principle that rules may not refer to the absence of a feature. It is also discussed by Lightner (1971:236). Hayes' (1986:331) Linking Constraint ("association lines in structural descriptions are interpreted as exhaustive") is an attempt to address this problem in autosegmental phonology; see also discussion by Kiparsky (1985) and Archangeli (1988). Further problems and a proposed solution are discussed at length by Reiss (2003). Bonet (1995:631, fn. 28) notes the same logical issue in an analysis of clitic selection, but does not offer a solution.

The approach taken in LP, first proposed by Bale et al. 2014, is to introduce changes in the featural content of segments via *unification*, denoted by the $\sqcup$ operator. The unification of a segment (set) and the set of features corresponding to the structural change is equivalent to the set theoretic *union* of the two sets, but only if the union of the two sets is consistent, if it contains no conflicting feature values, like +F and −F. If the union of the two sets is not consistent, unification *fails* (or equivalently, its output is *undefined*). Unlike union, which always yields an output set for any two input sets, and is called a total operation, unification is a partial operation, a property that LP exploits.

A rule that unifies the segment /i/ with the set {+HIGH} applies vacuously (i.e., has no effect), because the set denoted by /i/ already contains +HIGH, so unification is vacuous. However, the same rule applied to the segment /e/, specified as −HIGH, results in unification failure because the *union* of the two sets includes both +HIGH and −HIGH and is thus inconsistent. We assume, as part of the overall interpretative procedure for rule application, that when unification fails in a phonological rule application, the target segment (the first of the two arguments) remains unchanged.[3] Thus, the unification *rule* applied to /e/ with {+HIGH} vacuously yields /e/. Finally, the unification of /I/ with {+HIGH} is equivalent to the union of /I/ with {+HIGH}, and yields /i/. The rule *appears* to have only targeted /I/, but this is an illusion: it non-vacuously targets /I/ and vacuously targets /i, e/, though for different reasons in each case.

Reiss (2021) gives copious examples of this mechanism, as well as a second type of rule involving set difference or set subtraction, denoted here by the $\setminus$ operator. Subtraction followed by unification yields feature-changing processes. Subtraction can remove features and unification can add them. As in ordinary set theory, $A \setminus B$ denotes a set containing all and only those elements

[3] Unification is modeled on a similar operation used in various syntactic formalisms (e.g., Shieber 1986); the convention of interpreting unification failure as vacuous application is also used in that literature.

of A which are not elements of B; for example, $\{a, b, c\} \setminus \{a, x, y\} = \{b, c\}$. Subtraction can also be vacuous, as in $\{a, b, c\} \setminus \{x, y\} = \{a, b, c\}$. Together, rules based on unification and subtraction cover a wide range of phonological patterns, and eliminate the need for rule diacritics distinguishing between feature-filling and feature-changing rules.

Below we put this feature logic to use in the study of metaphony, but we note, once again, that metaphony is a pre-theoretic, phenomenological label that has no formal status in our theory. Our LP assumptions will allow us to develop the following four basic themes.

(8) Themes of the LP approach to metaphony:

i. Apparent lexical and/or morphosyntactic conditioning of phonological targets can be handled by a judicious choice of underlying forms, especially the use of underspecification.

ii. Apparent lexical and/or morphosyntactic conditioning of phonological environments can be handled with underspecification along similar lines.

iii. Unification provides an elegant solution to the formalization of feature-filling processes without the need for rule diacritics.

iv. Feature-changing processes reflect subtraction followed by unification.

These notions are all present, in some form or another, in the literature, but LP provides a simple unifying framework to express them, and metaphony provides the corpus with which we illustrate them. Before proceeding, we note that some patterns previously described as metaphony reflect completed sound changes that correspond to no synchronic rule. LP is a theory of synchronic grammar and thus it has nothing to say about language change.

2.4 Favoring narrow phonology

For examples of metaphony that are potentially amenable to synchronic analysis, some, classified as *morphophonological*, appear to require rules or constraints referring to the identity of particular morphemes—i.e., those which undergo metaphonic alternations—and/or to morphosyntactic contexts where those alternations occur. Other productive cases of metaphony can be analyzed straightforwardly as purely phonological processes. For clarity we refer to this latter type of metaphony as generated by the *narrow phonology*. While we provide a detailed comparison of morphonological and narrow phonological analysis in section 4, our goal in this paper is to show that our restrictive model of the narrow phonology can handle at least some metaphony patterns that might otherwise be attributed to morphophonology.

We adopt the standard—but usually implicit—position that a narrow phonological analysis is preferable, *ceteris paribus*, to a morphophonological analysis. We interpret this *ceteris paribus* condition to mean that a purely phonological solution is to be favored if it can be achieved without enriching the ontological commitments of the narrow phonology, by exploiting only a restricted, independently motivated phonological toolkit. Under this strategy, it is desirable to attempt to bring all instances of synchronic metaphony under the umbrella of the narrow phonology. We hypothesise that children are *epistemically bounded* (in the sense of Fodor 1980:33f.) to pursue narrow phonological analyses before entertaining morphemic or morphosyntactic conditioning. Only when the innately determined representational and computational resources of the narrow phonology fail to generate such a one-to-one mapping does the learner resort to morphophonology. Of course, the child has an advantage over the analyst: the child knows what phonological Universal Grammar consists of. We, the linguists, in contrast, have to rely on Ockham's Razor and adopt an initially restrictive model.[4] This position—that phonology defines the extent to

[4]This demonstrates the problem with the application of what Gopnik (2003) calls "theory theory" to cognitive development. "Theory theory" proposes a strong parallel between the learning that takes place in development and the work of scientists. However, scientists can run experiments and make use of negative evidence, which children are believed to largely ignore; and children's genetic endowment constrains the hypothesis space for language, whereas scientists have to discover that, say, syntactic processes require reference to novel notions like c-command.

which the one-to-one correspondence holds between sound and meaning—may not be correct, but it is at least an explicit hypothesis that can be explored, and it provides a principled justification for prioritizing phonological solutions rather than delegating these problems to morphology. As we show in section 4, even architectural theories with a a highly-articulated morphophonological component stand to benefit from our approach, since we show that some of the patterns for which they have accepted responsibility can, in fact, be reassigned to the narrow phonology under our assumptions.

2.5 Inalterability as prespecification

We take inspiration from work by Inkelas and colleagues (e.g., Inkelas and Cho 1993, Inkelas 1995, Inkelas and Orgun 1995, Inkelas et al. 1997) on apparent exceptions to phonological processes analyzed under the slogan *Inalterability as Prespecification* (IAP). In brief, the idea is that given a feature-filling rule that supplies segments with a specification of +F, the rule will modify segments lacking a specification for F. However, segments that appear immune to the rule—i.e., that are not altered by the rule—should be understood as "prespecified" as +F or −F. In some sense, these prespecified segments are not exceptional; they just are not subject to change by a feature-filling rule. As discussed above, unification—and its vacuous application via unification failure—provides a formal mechanism to implement feature filling. If the tools of narrow phonology include operations like unification and subtraction, and if underspecified segments are permitted, then we predict a child will converge on grammars like those we propose below.

3 Baztan low vowel assimilation

Before turning to a more traditional case of metaphony, let us first consider an example of vowel-vowel interaction to illustrate the IAP insight. Hualde (1991) discusses a process called Low Vowel Assimilation (henceforth, LVA) present in several dialects of Basque. LVA is attributed to "a rule that raises a low vowel to /e/ after a high vowel, with or without any intervening consonants" (op. cit.:23). Some examples, common to all dialects with LVA, are given below; /-a/ is the singular definite suffix.

(9) Low Vowel Assimilation (loc. cit.):

/mutil-a/	[multil**e**]	'the boy'
/mendi-a/	[mendi**e**]	'the mountain'
/egun-a/	[eɣun**e**]	'the day'

Our discussion here focuses on Hualde's account of the Baztan dialect (§2.2). In this dialect, which has five surface monophthongs [a, e, i, o, u] as well as a number of diphthongs, the process affects vowels within words and across certain word boundaries, as shown in the following examples.

(10) Auxiliary verbs with raised vowel after high vowel (op. cit.:29–30.):

a.	torri d**e**	'he has come'	cf. gan d**a**	'he has gone'
b.	gain d**e**	'he will go'	cf. torriko d**a**	'he will come'
c.	torri g**e**ra	'we have come'	cf. gan g**a**ra	'we have gone'
d.	torri z**e**ra	'you have come'	cf. gan z**a**ra	'you have gone'
e.	in z**e**zu	'do it'	cf. jan z**a**zu	'eat it'

In (10a), for example, the auxiliary *de* has an [e] mid vowel after the verb *torri*, ending in [i]. In contrast, the same auxiliary has a low [a] after the verb *gan*, whose last vowel is [a]. Similar patterns are found in other examples.

Hualde discusses this data in a chapter where his goal is to "explore the interaction between the phonological rules of Basque and different morphological operations, with the purpose of determining the role of the morphology in defining the domain of application of phonological processes" (op. cit.:22). Hualde claims that morphemes in which /a/ is unaffected belong to different morphological strata than those which undergo LVA. However, it seems difficult to maintain this position given his characterization of the data.

(11) Exceptions to LVA (op. cit.:26–30):

- "Not all derivational suffixes present the same behaviour with respect to Low Vowel Assimilation."
- "Verbal suffixes generally undergo assimilation."
- "We need to examine now the application of the rule in conjugated verbal forms. Here, the situation is not uniform… [I]n quite a few conjugated forms the rule fails to apply."
- "The forms that present the context for Low Vowel Assimilation, but, nevertheless, do not undergo the rule, have just one more irregularity that must be lexically marked."
- "Certain auxiliary verbs also undergo assimilation… Other forms of the auxiliary, on the other hand, never undergo assimilation, including the other persons of the intransitive present indicative not mentioned above…"
- "We must conclude that only a few auxiliary forms can behave like clitics and thus undergo Low Vowel Assimilation."

Our analysis of LVA is far simpler than Hualde's account of the exceptional morphemes, given the large number of exceptional morphemes. For example, in contrast with (10), there are other auxiliary verb forms that do not exhibit LVA. The 1sg. *naiz*, the 2sg. informal *aiz*, and the 2pl. form *zate* are all invariant, as seen in *torri naiz* 'I have come'. It is hard to imagine any account which would place non-alternating auxiliaries like *naiz* and alternating auxiliaries like those in (10) in different morphological strata. We propose that Baztan LVA is best analyzed as an instance of IAP. We posit an underlying three-way distinction between /a, e, A/, in which /A/ is −Back and −High but unspecified for Low, as schematised below.[5]

[5] For concreteness, we follow Hualde (op. cit.:197, en. 2) in taking /a/ to be −Back.

(12) Partial feature specification for Baztan:

	/a/	/A/	/e/
Back	–	–	–
High	–	–	–
Low	+		–

In our analysis, it is not the case that the different auxiliary forms are assigned to different morphological strata; rather, some of them contain an underspecified /A/ which, because of its underspecification, is *mutable* (with respect to a unification rule), and others contain a fully-specified /a/ which is *inalterable* (with respect to this rule) because its richer specification gives rise to unification failure. This rule is given below; note that the word boundary is not indicated.

(13) $\left[-\textsc{High}\right] \sqcup \left\{-\textsc{Low}\right\} \;/\; \left[+\textsc{High}\right]\, C_0$ —

The relevant portion of the Baztan phoneme inventory are the –High vowels /a, e, A, o/. Unification of /a/ with {–Low} is vacuous because this vowel's +Low specification causes unification failure. Unification of /e, o/ with {–Low} is also vacuous because these vowels are already specified –Low. Only unification with /A/ is non-vacuous and produces raising. Crucially, though, rule (13) *appears* to have targeted only the underspecified segment even though LP's notion of natural classes and feature-filling make it impossible to do so, since the smallest natural class containing /A/ contains all four –High vowels /a, e, A, o/. The rule intensionally targets all –Low vowels, but the only vowel affected by it is /A/.

Now, it is also necessary to ensure that tokens of /A/ which do not undergo (13) surface as the +Low vowel [a]. This can be done by the following context-free redundancy rule, ordered after (13). This maps the remaining /A/s to [a], vacuously applying to all other segments.

(14) $\left[-\textsc{High}\right] \sqcup \left\{+\textsc{Low}\right\}$

This example illustrates themes (8i) and (8iii). Hualde's appeal to morphological strata to generate exceptions to LVA seems unworkable, and indeed, he provides no means by which to stratify morphemes except by considering whether or not they exhibit LVA. We avoid this by positing an underspecified vowel /A/, by its nature mutable and thus a potential target for LVA, and contrast this with pre-specified, inalterable /a/, which is immune to LVA. Unification and its failure implement a feature-filling rule without the need for rule diacritics or morphological strata. LP's adoption of IAP means that there are no actual exceptions to LVA in the narrow phonology of Baztan.

4 Spanish third conjugation metaphony

We now turn to an example that has traditionally been referred to as metaphony (Malkiel 1966), the so-called "raising" alternations between [e] and [i] in Spanish verbs of the third conjugation, as in *p**e**dí* 'I asked' vs. *p**i**do* 'I ask' or *g**e**mí* 'I wailed' vs. *g**i**mo* 'I wail'. There are over one hundred raising verb stems, though most of these stems are prefixed forms of a few dozen distinct roots; for example, the raising verb *decir* 'to say' has many prefixal variants including *antedecir* 'to foretell', *maldecir* 'to curse', *redecir* 'to retell', and so on. Many other third conjugation verbs, like *vivir* 'to live' and *sumergir* 'to submerge' do not alternate—the former has invariant [i] and the latter has invariant [e].[6] One could imagine analyzing the raising verbs as involving suppletion between listed allomorphs (e.g., /ped-/ vs. /pid-/), but as Embick (2012:33) notes, such an analysis would have to make reference to a complex, highly disjunctive set of morphosyntactic contexts.

[6] Spanish has five surface monophthongs—[i, e, a, o, u]—as well as several diphthongs. A few raising verbs (e.g., *m**e**ntí* 'I lied' vs. *m**i**ntió* 's/he lied') also participate in alternations in which stem stress triggers diphthongization (e.g., *m*[je]nto 'I am lying'); see Embick 2012:§3.1 for discussion.

(15) Morphosyntactic contexts for /ped-, pid-/:

a. /ped-/: 1pl./2pl. present indicatives, 1sg./1pl./2sg./2pl. preterites, all imperfectives, all futures, all conditionals

b. /pid-/: 1sg./2sg./3sg./3pl. present indicatives, all present subjunctives, all imperfect subjunctives, 3sg./pl. preterites

There is no obvious way to treat the difference between (15a) and (15b) in terms of natural classes based on morphosyntactic features, so the grammar would necessarily contain two lists—a list of which verb *roots* participate in the alternation, and a list of which verb *forms* on this preceding list take [i] vs. [e].[7] Embick concludes that "[a]n analysis that makes reference to morphosyntactic features thus looks very unpromising", and he offers a morphophonological alternative. He notes that [e] alternants of raising verbs are found when the following syllable contains an [i], and [i] alternants are found in all other cases. Following a suggestion of Harris (1969:§4.1.2.2), Embick takes the alternating vowel to be underlying /i/, and proposes the following rule to generate the [e] allomorphs.

(16) $\mathrm{i} \rightarrow \mathrm{e}$ / __ $\mathrm{C_0 i}$ (condition: certain roots)

Embick's proposal improves on the morphosyntactic analysis, because it requires only a single list of which verb roots are included in (16) and not a list of specific forms, as in (15): this "(morpho)phonological analysis of the Raising verbs is straight-forward, as long as it is acknowledged that the phonological process can be Root-specific." Note that Embick has replaced the traditional "raising" with a rule of lowering, since he has underlying /i/ surface as [e]. Given our adherence to the substance-freeness of structural change (2), we have nothing to say about his use of a dissimilatory lowering rule rather than an assimilatory raising rule.

[7] Embick (op. cit.) also notes that a stem suppletion account of raising would run counter to his theory of locality conditions on stem suppletion (Embick 2010).

However, with our LP technology, we can go a step further than Embick and replace his rule (16) with a narrow phonological rule. We can provide an account that reduces all idiosyncracy to the phonological representation of individual morphemes in the lexicon, the level of the grammar which is characterised by idiosyncratic representations of contrastive elements. Suppose that the alternating vowel in raising verb stems is underlyingly /I/, with the same featural specification as in (4)—i.e., it is a vowel underspecified with respect to High but otherwise the same as non-alternating /i, e/. Then instead of a rule like (16), we can provide a purely phonological rule that makes no reference to the identity of particular verb roots.

(17) $$\begin{bmatrix} -\textsc{Back} \\ -\textsc{Low} \end{bmatrix} \sqcup \left\{ -\textsc{High} \right\} \;/\; \_\_\; C_0 \begin{bmatrix} -\textsc{Back} \\ +\textsc{High} \end{bmatrix}$$

This rule maps underspecified /I/ to [e] when there is an /i/ in the next syllable; it has no effect on fully-specified /i, e/.

To derive words like *pido*, in which /I/ is not followed by /i/ and thus not targeted by (17), we posit the following context-free rule, which maps remaining /I/ to /i/ which applies vacuously to fully-specified /e, i/.

(18) $$\begin{bmatrix} -\textsc{Low} \end{bmatrix} \sqcup \left\{ +\textsc{High} \right\}$$

We note that the raising verbs are not the only evidence for /I/ in Spanish. As is well-known (e.g., Mascaró 2007:722), the coordinating conjunction *y* 'and', pronounced [i], is realized as *e* [e] when the following word begins with [i], as *guapa e interesante* 'pretty and interesting'. Assuming that the conjunction is underlyingly /I/, the relevant rule is quite similar to (17).

(19) $$\begin{bmatrix} -\textsc{Back} \\ -\textsc{Low} \end{bmatrix} \sqcup \left\{ -\textsc{High} \right\} \;/\; \_\_\; \# \begin{bmatrix} -\textsc{Back} \\ +\textsc{High} \end{bmatrix}$$

Ordering this rule before redundancy rule (18) derives the expected pattern.

This analysis again illustrates themes (8i) and (8iii), and it is quite similar in spirit to earlier IAP analyses by Inkelas and colleagues and to our analysis of Baztan. Just as in Baztan, where unraised

/A/ surfaces as /a/, it is necessary to ensure in Spanish that underspecified /I/ vowels which do not undergo lowering by (17) surface as [i] and not [e]. This poses no particular difficulty given the default rule (18). Our contribution is to show that the operations of LP, combined with IAP, allow us to treat Spanish raising verb allomorphy in the narrow phonology. Crucially, our purely phonological solution does not exact any price: we only use tools independently proposed to address unrelated phenomena. We can account for Spanish raising verbs in the narrow phonology without enriching our model.

5 Cervara feature-changing metaphony

Metaphonic patterns cannot all be framed as feature-filling, however, so we now provide an analysis that incorporates a subtraction rule. A particularly clear case is apparent in Cervara, a dialect spoken in Lazio documented by Clemente (1922) and analyzed by Danesi (2022:§7.2.2). We present a simplified version of Danesi's data focusing on front vowel targets to illustrate salient properties of the LP approach.

5.1 Metaphony in verbs

We first consider metaphony in Cervara verbs. Like Standard Italian, Cervara has seven surface monophthongs [a, ɛ, e, i, ɔ, o, u]. At first glance it appears that the mid vowels /ɛ, e/ raise and neutralise to high, tense [i] when the final vowel is high, tense /i, u/.

(20) Cervara verbs (op. cit.:514):

a.	ˈv**e**do	'I see'	ˈv**i**di	'you see'
	ˈv**e**de	'he sees'	ˈv**i**du	'they see'
b.	ˈm**ɛ**to	'I reap'	ˈm**i**ti	'you reap'
	ˈm**ɛ**te	'he reaps'	ˈm**i**tu	'they reap'

Ignoring ROUND and other irrelevant features, we hypothesise that the vowels which surface as [e] in "unraised" form, are underlyingly /e/, specified as {−LOW, −HIGH, +ATR}, whereas those that surface as [ɛ] in unraised form are underlyingly {−LOW, −HIGH} but underspecified for ATR, a vowel we denote as /E/. We first propose the following rule which changes /E/ to [e] before a tense, high vowel, /i/ or /u/.

(21) $\begin{bmatrix} -\textsc{Low} \\ -\textsc{High} \end{bmatrix} \sqcup \left\{+\textsc{Atr}\right\} \;/\; \_\_ \; C_0 \begin{bmatrix} +\textsc{High} \\ +\textsc{Atr} \end{bmatrix}$

This is yet another feature-filling rule implemented with unification—like (13) or (17)—that neutralises the contrast between the two mid vowels by adding a +ATR specification in a particular context. This rule feeds the next step of the process, which raises mid vowels specified +ATR if they are followed by a +HIGH vowel. This is a feature-changing process, because the target class includes vowels that are specified −HIGH. In LP, feature changing is a two-step process, with subtraction followed by unification. We first need a rule to remove the −HIGH specification from the relevant mid vowels.

(22) $\begin{bmatrix} -\textsc{Low} \\ +\textsc{Atr} \end{bmatrix} \setminus \left\{-\textsc{High}\right\} \;/\; \_\_ \; C_0 \begin{bmatrix} +\textsc{High} \end{bmatrix}$

This rule then feeds a third rule which raises +ATR vowels when followed by a +HIGH vowel.

(23) $\begin{bmatrix} -\textsc{Low} \\ +\textsc{Atr} \end{bmatrix} \sqcup \left\{+\textsc{High}\right\} \;/\; \_\_ \; C_0 \begin{bmatrix} +\textsc{High} \end{bmatrix}$

The feature-changing conversion of mid to high vowels is thus accomplished by a subtraction rule (22) followed by a unification rule (23), illustrating theme (8iv). Some sample derivations are provided below.

(24) Sample Cervara derivations (stress omitted):

UR	ved-i	mEt-i
Rule (21):		meti
Rule (22):	vIdi	mIdi
Rule (23):	vidi	midi
SR	vidi	midi

5.2 Metaphony in nominals

There is more to metaphony in Cervara. A different pattern is found in nouns and adjectives (henceforth, nominals). In nominals, the underlyingly tense /e/ vowel raises to [i] when followed by a high vowel, just as we saw in verbs in (20). However, unlike the situation in verbs, in nominals underlying /E/ just shifts to /e/, surfacing as +Atr but remaining mid.

(25) Cervara nominals (op. cit.:508–509):

a.	fr**e**lˈl**e**nka	‘vulva (vulgar)’	fr**i**lˈl**i**nku	‘penis (vulgar)’
b.	ˈbːɛlle	‘beautiful (fem. pl.)’	ˈbː**e**ʎʎi	‘beautiful (masc. pl.)’

The solution we propose is essentially a translation of Danesi’s proposal into LP. Our analysis reflects theme (8i) in that it provides a narrow phonological solution for allomorphy that superficially appears to require morphological conditioning, and it illustrates theme (8ii) in that it involves an extension of IAP from targets to triggers. We distinguish between *catalytic* segments which trigger the alternation and surface-identical *quiescent* segments which do not.

This extension of IAP appears implicitly in the literature,[8] but we suspect that the implications have been insufficiently appreciated because the approach is so simple. The IAP distinction

[8] For example, an anonymous reviewer draws a connection between our analysis of Cervara and Inuit dialects said to have “strong” and “weak” *i*’s (e.g., Compton and Dresher 2011). While we do not necessarily endorse Compton and Dresher’s theory of contrastive specification, we note that their assumptions lead them to posit a “strong” and “weak” *i*’s that parallel the Cervara segments /i, Y/ defined below.

between two types of potential targets via underspecification requires no enhancements to phonological theory beyond admitting underspecification, something Inkelas and colleagues argue for in detail.

Therefore, it is important to note that this same idea can also be extended to differentiate segments that are potential rule *triggers*, a possibility not exploited by Inkelas and colleagues. Danesi's analysis of Cervara (§7.2.2) makes use of this idea to avoid the need for lexical or morphosyntactic conditioning on metaphonic triggers. Our proposal is a translation of Danesi's analysis in LP. However, it demonstrates the possibility of analyzing the complex Cervara pattern with rather different theoretical primitives while preserving Danesi's insightful hypothesis that the relevant metaphonic patterns can be encoded within the narrow phonology.

Once again, for ease of exposition, we restrict our attention only to front vowel targets. Back vowel behavior is completely symmetrical, so we set aside Back and Round in our discussion. As above, we hypothesise that vowels which are realised as [e] in "unraised" form are {−Low, −High, +Atr} and those that are realised as [ɛ] in unraised form are {−Low, −High} but not specified for Atr, so /E/. Consequently, there exists a natural class containing just /e/ using [−Low, −High, +Atr], but there is no corresponding singleton natural class for /E/: the class [−Low, −High] also contains /e/. This is a desirable result.

Danesi presents an analysis of Cervara with two distinct sets of high vowels. One set, found in nominal endings, triggers tensing of [E] but raising of [e]. The other set triggers tensing and raising of /E/, and raising of both underlying and derived +Atr [e]. We propose that the two triggering vowels, both of which are realised as [i], are underlyingly distinct, so this is a case of *absolute neutralization* in the sense of Kiparsky (1968). The 2sg. verb ending /-i/ is specified [−Low, +High, +Atr]. The nominal ending, which triggers the less drastic alternations, is denoted by /Y/ and is specified [−Low, +High], without a specification for Atr.[9] By the same

[9]Note that we depart from Danesi's notation to follow the traditional convention of denoting underspecified segments with capital letters.

reasoning given above, there exists a singleton natural class containing only /i/, but the smallest natural class containing /Y/ also contains /i/.

In nominals, −Atr mid vowels change to +Atr in the presence of / Y/, while +Atr vowels become high vowels. Therefore, it is crucial to order the /E/-to-[e] rule after the /e/-to-[i] rule, in counterfeeding order. Recall that in verbal forms, both mid vowels raise to [i]. We now incorporate the rules developed above for verbs into an account that also covers nominals; note that schematic derivations are again provided once the rules are laid out.

The first step is rule (21), repeated below as (26). This rule neutralises the distinction between /e, E/ in the presence of a following /i/. Crucially, the +Atr environment specification ensures that /Y/ does not trigger this rule.

(26) $\begin{bmatrix} -\textsc{Low} \\ -\textsc{High} \end{bmatrix} \sqcup \left\{+\textsc{Atr}\right\} / \_\_ \mathrm{C}_0 \begin{bmatrix} +\textsc{High} \\ +\textsc{Atr} \end{bmatrix}$

The next step raises all +Atr mid vowels followed by a +High vowel. This will target underlying /e…i, e…Y, E…i/ but not /E…Y/ since /Y/ is unspecified for Atr and thus will not have triggered the preceding rule. This is a feature-changing process, and is accomplished by a subtraction rule, repeated from (22), followed by a unfication rule, repeated from (23).

(27) $\begin{bmatrix} -\textsc{Low} \\ +\textsc{Atr} \end{bmatrix} \setminus \left\{-\textsc{High}\right\} / \_\_ \mathrm{C}_0 \begin{bmatrix} +\textsc{High} \end{bmatrix}$

(28) $\begin{bmatrix} -\textsc{Low} \\ +\textsc{Atr} \end{bmatrix} \sqcup \left\{+\textsc{High}\right\} / \_\_ \mathrm{C}_0 \begin{bmatrix} +\textsc{High} \end{bmatrix}$

Then, underlying /E/ before /Y/ is specified +Atr.

(29) $\begin{bmatrix} -\textsc{Low} \end{bmatrix} \sqcup \left\{+\textsc{Atr}\right\} / \_\_ \mathrm{C}_0 \begin{bmatrix} +\textsc{High} \end{bmatrix}$

Note that the target of (29) is the natural class characterised intensionally so as to include /e, E/, but in practice it only applies to /E/ because an /e/ followed by /i, Y/ will already have been raised by (28). Similarly, the trigger for (29) intensionally includes /i, Y/, but it only applies to /E...Y/ sequences simply because there are no more /E...i/ sequences at this stage of the derivation thanks to rules (26–28).

The final two rules are redundancy rules. Such rules have no special status in LP: they are simply context-free unification rules, like the ones we saw in Baztan and Spanish. The first rule introduces a +Atr specification to high vowels not specified for Atr. While this is only needed for /Y/, recall that our logic requires that we intensionally target /i, Y/, even though unification applies vacuously to the former. It is also important to note that a vowel like /Y/ lacks an Atr specification but may still trigger insertion of +Atr given the substance freeness of structural change (2). LP does not require a copy of a feature inserted or deleted by a rule to be present in that rule's structural environment. The second rule introduces a −Atr specification for mid vowels not specified for Atr, causing remaining tokens of /E/ to surface as [ɛ].

(30) $\left[+\textsc{High}\right] \sqcup \left\{+\textsc{Atr}\right\}$

(31) $\left[-\textsc{High}\right] \sqcup \left\{-\textsc{Atr}\right\}$

These last two rules can be expressed as context-free because at this stage of the derivation, only a subset of vowels lack an Atr specification. The rules apply vacuously to fully specified vowels.

The following example shows the schematic derivation of the relevant sequences. We use /I/, as in Spanish, to denote a vowel—appearing only in intermediate levels of derivation in Cervara—specified +Atr but underspecified with respect to High. The last two columns provide derivations for the mid vowels not occurring in the contexts for metaphonic alternations.

(32) Sample Cervara derivations (stress omitted):

	vrb.	nom.	vrb.	nom.		
UR	e…i	e…Y	E…i	E…Y	e	E
Rule (26):			e…i			
Rule (27):	I…i	I…Y	I…i			
Rule (28):	i…i	i…Y	i…i			
Rule (29):				e…Y		
Rule (30):		i…i		e…i		
Rule (31):						ɛ
SR	i…i	i…i	i…i	e…i	e	ɛ

This analysis of Cervara could be adapted to similar cases of metaphony, for example, in the northern Salentino variety of Francavilla Fontana discussed by Calabrese (1985) and Torres-Tamarit and Linke (2016). Historical vowel mergers make this pattern a challenge for traditional phonological analysis. In adjectives, for example, the [-i] of the masculine plural and the [-u] of the masculine singular appear with a raised stem vowel, but the [-i] of the feminine plural does not correlate with raising of the preceding mid +Atr vowel.

(33) Francavilla Fontana Salentino adjectives (Calabrese 1985:11):

	sg.	pl.	
fem.	ˈfredda	ˈfr**e**ddi	‘cold’
masc.	ˈfr**i**ddu	ˈfr**i**ddi	

A similar analysis to the one we provide for Cervara is feasible here. Apparently, the underlyingly distinct affixes in this variety of Salentino—by hypothesis, fully-specified masc. pl. /-i/ and underspecified fem. pl. /-Y/—express different gender features rather than different syntactic categories. In LP, it is not only possible to posit different phonological features for the two plural endings, but it is also unclear how LP might be constrained in a nonstipulative manner to preclude analyses of this type.

6 Conclusion

We have attempted to provide a consistent feature logic under an explicit set of assumptions, such as the use of binary features, the interpretation of segments and natural classes as sets, and the possibility of underspecification. Many open questions remain, and alternative analyses are available. For example, one might posit that the Cervara phoneme denoted by /E/ lacks, say, a Spread Glottis specification rather than an Atr specification. Furthermore, some phonologists and morphologists may be skeptical of our use of absolute neutralization. We present two arguments against such skepticism.

We call the first argument the "black hole" argument. Black holes cannot be directly observed since no information can escape them. Their existence can only be discerned from their effect on other objects. Since our proposals for distinct phonological entities, for example the two underlying sources of [i] in Cervara, are based on their differential effects on other elements (for example, to what degree they mutate a nearby /E/), there must be some representational distinction at play. We are simply using the evidence at hand to infer a distinction, and the tools on hand to encode it, and it is conceivable that our hypotheses could someday be confirmed or falsified by behavioral or neuro-imaging evidence.

The second argument is of a more general kind. It appears to be the case that a theory that makes use of ordered rules, binary features and underspecification, unification and subtraction, and the principle of substance-freeness of structural change (2) has no obvious way to exclude the sort of analyses we have proposed. We do not assert that the proposals here are true and correct, but simply that it is difficult to demonstrate they are invalid, since the representations and operations they are built out of are all themselves well-supported. Our analyses follow from the components of our theory, which has no way to rule out a phonological system like Cervara with two underlying sources for [i]. Thus absolute neutralization is not only not forbidden, it is expected. Criticism of our assumptions and logic is of course quite welcome, but the overall system is fairly simple and yields rich empirical coverage via its combinatorics. A simple model generating vast phenomenological complexity is exactly what we want out of a scientific theory:

> As concepts and principles become simpler, argument and inference tend to become more complex—a consequence that is naturally very much to be welcomed. (Chomsky 1982:3)

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